SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 3

Design of Products and Services

Design of products and services (Ch. 3) with two HBS cases: Apple (Design Thinking & Innovation, 9–609–066) and Shell (A Company of Opportunity?, 9–320–025). Covered: why design is an operations decision, firm boundaries (vertical integration to pure marketer, core technological competency), the six generic development phases and seven variants, platform products, design criteria (QFD & House of Quality, DFMA & concurrent engineering, VA/VE, ecodesign), service design (three factors, complexity & divergence), new-product economics (NPV, sensitivity, development metrics), Design Thinking and SCAMPER, and seven discussion questions.

Material

1 Learning objectives and roadmap

1. Evaluate the economic impact of a new product and how development is measured.
2. Know product design issues and typical processes.
3. Illustrate how criteria shape a design.
4. Contrast service and manufactured product design criteria.

Foundations → Process: 6 phases, 7 variants

→ Criteria: QFD, HoQ, DFMA, VA/VE, ecodesign → Services & economics

→ Design Thinking & SCAMPER → Apple & Shell

2 Why product design is an operations problem

- Design locks in most of a product's lifetime cost before the first unit.
- It determines what operations must be capable of — process, suppliers, quality, capacity.
- It expresses strategy: what you include and deliberately leave out.
- In fast markets, the ability to redesign repeatedly matters more than one perfect design.

Simplicity
Not the same as stripped down.

Sophistication
NeXT kept features rivals lacked for years.

Discipline
Saying no to a thousand ideas.

Boundary	Meaning	Example
Fully vertically integrated	Everything in-house	—
Design in, build out	Design and keep IP, subcontract fabrication	Apple iPhone
Orchestrated network	Buy/contract most stages, keep patents	Pharma
Pure marketer	Only sells	—

Core technological competency
Black & Decker builds on 200–600 W motors across workshop, cleaning and kitchen markets.

3 The six phases of generic development

Phase	Name	Content
0	Planning	From corporate strategy; technology and market objectives; output: mission statement.

1	Concept development	Needs identified; concepts generated, evaluated, selected with specs and economic justification.
2	System-level design	Architecture, subsystems, geometric layout, preliminary assembly flow.
3	Detail design	Geometry, materials, tolerances; process plan; tooling.
4	Testing & refinement	Preproduction versions tested against design and customer needs.
5	Production ramp-up	Using the real production system to train and debug, then launch.

This describes a **market-pull** situation.

4 Seven process variants and platform products

Type	Feature	Examples
Technology-push	New proprietary technology seeks a market	Gore-Tex, Tyvek
Platform	Built on an established subsystem	Consumer electronics, iOS, Prius
Process-intensive	Product constrained by process	Semiconductors, snacks, chemicals
Customized	Slight variations	Motors, switches, batteries
High-risk	High chance of failure	Pharma, space
Quick-build	Many design–build–test cycles	Software, phones
Complex systems	Many subsystems in parallel	Airplanes, jet engines, cars

Platform products: plan the family early; the first product is the platform. Benefits: investment leveraged across derivatives, shared parts (reliability,

cost), knowledge carried over, lower customer training. Apple: OS X/NeXTStep, the iPod family, iPhone → iPod Touch → iPad.

5 Designing for the customer: QFD and the House of Quality

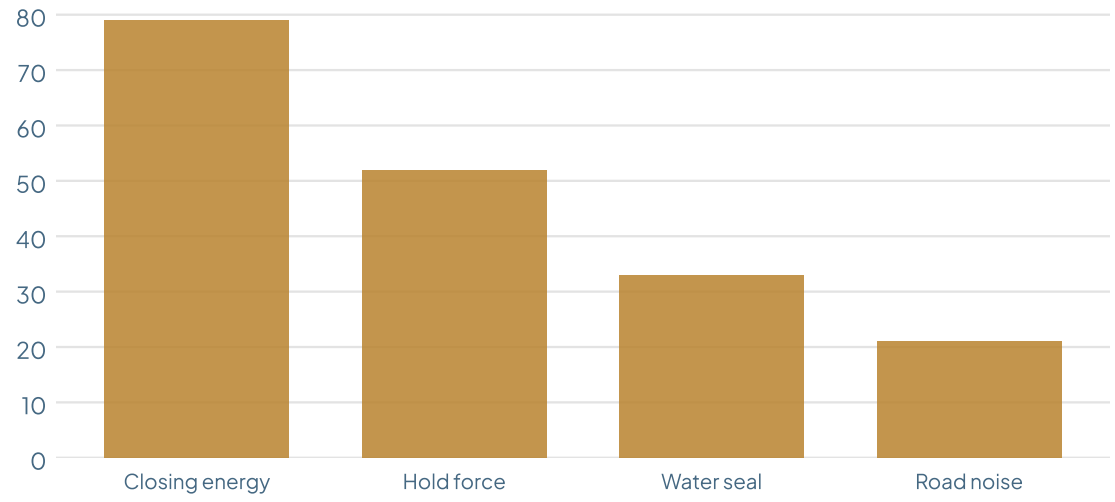
QFD brings the voice of the customer into the specification — Toyota credits it with cutting car costs by over 60% through shorter design times.

1. Listen
2. Categorise into customer requirements
3. Weight by importance
4. Benchmark against competitors
5. Translate in the House of Quality

> **House of Quality**
> A cross-functional matrix translating ranked requirements into engineering goals; it forces marketing, design and manufacturing to agree.

Requirement	Weight	Closing energy	Hold force on hill	Water seal	Road noise
Easy to close from outside	7	●●●	●		
Stays open on a hill	5	●	●●●		
No leaks in rain	3	●●		●●●	●
Quiet cabin	2	●		●●	●●●

Technical importance scores



Example scoring (weights × 9/3/1).

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Technical score = Σ (weight × strength); closing energy = 79
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6 DFMA, concurrent engineering, VA/VE and ecodesign

Over-the-wall problem

"We design it, you build it" caused late changes, expense and delay.

Concurrent engineering & DFMA: consult manufacturing during design; cross-functional teams develop product and process together; DFMA emphasises reducing production cost.

VA/VE: (1) unnecessary features? (2) combine parts? (3) cut weight? (4) eliminate nonstandard parts?

Ecodesign: environmental performance as a design requirement alongside quality, cost, manufacturability, function, durability, ergonomics, aesthetics. Principles: whole life cycle, product as a system, multicriteria. Benefits: producer (less material, water, energy, waste), user (reliable, durable, efficient, repairable), society (resources, avoided damage).

7 Designing service products: three factors, complexity & divergence

Direct customer involvement brings variability in service time and the knowledge employees need.

Factor	Question	Example
Similarity to current services	Fits the existing experience?	Disneyland photographers fit; a restaurant in a car wash doesn't.
Similarity to current processes	Is there operational support?	Grocery home delivery needed new skills.
Financial justification	Costly to design and run	Return may be profit or retention.

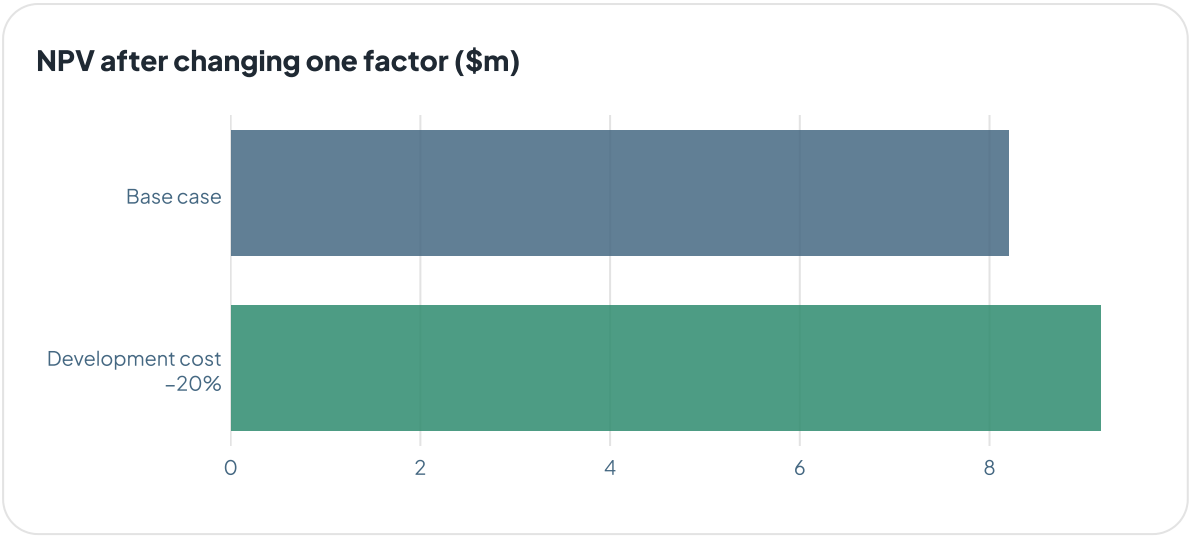
Complexity
Number of steps and possible actions at each.

Divergence
How much the interaction can vary at each step.

8 The economics of a new product: NPV and sensitivity

Build a base-case model of cash flows (development, ramp-up, marketing & support, production, sales) and compute NPV. Sensitivity changes one factor at a time:

Change	Impact
Development cost −20%	$1m less lifts NPV by $964,000 (+11.8%, $8.20m → $9.17m)
Development time +25%	4 → 5 quarters delays every later cash flow
Sales volume ±25%	The most powerful lever and the most exposed to forecast error
Price or unit cost $1	A dollar of profit — but price moves demand



Success metrics: time-to-market, productivity, quality (design and conformance, in DPMO).

9 Toolkits: Design Thinking (IDEO) and SCAMPER

IDEO sees overlapping spaces: inspiration, ideation, implementation.

1. Empathize
2. Define
3. Ideate
4. Prototype
5. Test & refine
6. Launch — and loop

Iterative
Tessler's participatory design; Atkinson rebuilt MacPaint from scratch; problems are moving targets.

SCAMPER	Meaning	Apple example
Substitute	Replace part/material/process	Aluminium iPod Mini

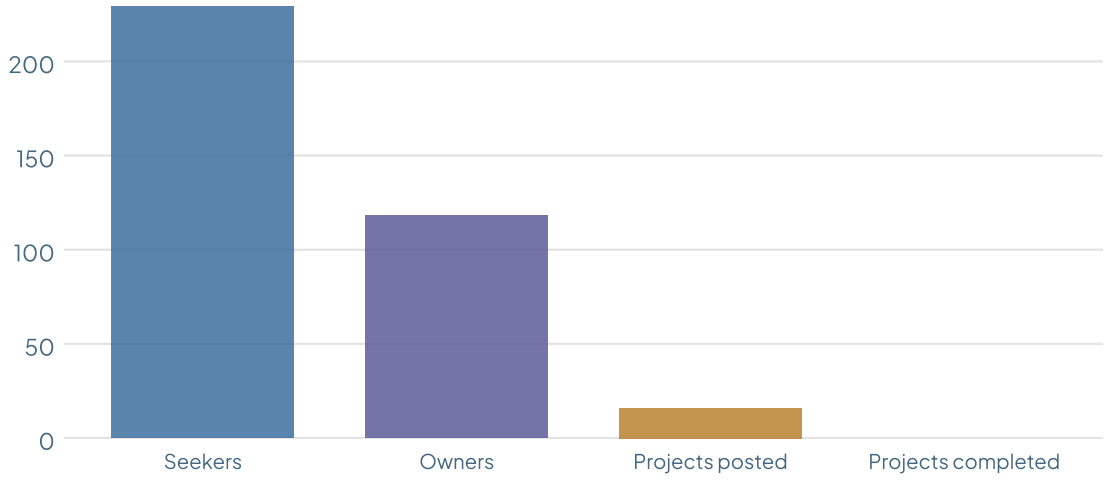
Combine	Join functions	iTunes + drive + store
Adapt	Fit a new context	OS X from NeXTStep
Modify	Resize/alter	Nano, Shuffle
Put to another use	New context	NeXT used to build the Web
Eliminate	Pare down	No diskette slot
Reverse	Invert	Bricks when everyone chose clicks

10 Case A (Apple) and Case B (Shell) in numbers

Apple (HBS 9-609-066): most valuable public company by March 2012, after ~$5 shares twelve years earlier. $600 share price; 25bn App Store downloads; 15 → 3 product lines after 1997 (70% of projects cut); 172m post-PC devices in 2011 (76% of Q4 revenue). The Apple Way: design thinking, platform strategy ruthlessly executed, CEO as chief innovator, bold experimentation (retail stores).

Shell (HBS 9-320-025): the Opportunity Hub let Owners post unstaffed projects and Seekers anywhere apply for up to 20% of their time. NPS 67; 118 Owners / 229 Seekers vs 16 projects (June 2018); 86,000 employees in 70+ countries; 40 unmatched Seekers still reported higher satisfaction.

Shell Opportunity Hub, June 2018



Worked

Reach beyond networks; Seekers wanted new skills; Owners got otherwise-outsourced resources.

Didn't

Vague briefs; line managers blocked release; scope creep vs day jobs.

11 Seven discussion questions (with answer directions)

Question	Answer direction
Q1 Why did removing features add value?	Simple to use ≠ removing function; Eliminate/Modify; destroys value if the removed feature is still widely needed.
Q2 What does the platform buy and risk?	Leverage, shared parts, knowledge, easy learning; risk when the core technology shifts; cannibalisation.
Q3 No market research yet user-driven?	Observation of real use vs QFD ranking; QFD for known products, observation for unimagined ones.
Q4 Retail stores: design or luck?	Apple designed a brand experience; Dell optimised cost and customisation; a bold but bounded experiment.
Q5 Diagnose Shell	Two customers; fails all three factors; variability enters through unstructured briefs and manager approval.

Q6 Redesign with SCAMPER	Structured brief template, link to appraisal, remove veto, smaller projects; measure completion and NPS.
Q7 Durable design capability	Principles embedded in systems, not one leader; Shell's structural constraints were never designed away.

12 Wrap-up: six takeaways

1. Design is an operations decision.
2. Adapt the process to the product.
3. Get the customer in early, by whatever method.
4. Services carry variability by design (complexity and divergence are choices).
5. Justify financially, then test the assumptions.
6. Toolkits turn a blank page into a starting point — not a substitute for understanding.

Key takeaways

1. Design locks in most lifetime cost and sets operational capability.
2. Firm boundaries: integrated, design in–build out, orchestrated network, pure marketer; core competency.
3. Six phases from planning to ramp-up (market-pull).
4. Seven variants: technology-push, platform, process-intensive, customized, high-risk, quick-build, complex systems.
5. Platform products leverage one architecture.
6. QFD and the House of Quality.
7. DFMA/concurrent engineering; VA/VE's four questions.
8. Ecodesign: life cycle, system, multicriteria.
9. Service design: three factors; complexity and divergence.
10. NPV and sensitivity; volume is the strongest lever.
11. Design Thinking and SCAMPER.
12. Apple's principles; Shell's demand-side failure.

Case studies

Case & worked answers

Apple — Design Thinking & Innovation

Apple went from ~$5 to $600 a share; 15 product lines cut to 3. Q1: removing features adds value when what goes is unvalued complexity (Eliminate/Modify) — it destroys value if the feature was a qualifier. Q2: platforms benefit firm, suppliers, employees and customers but risk obsolescence when the base technology shifts. Q3: observation and iterative prototyping are listening; QFD

suits known products, observation suits unimagined ones. Q4: Apple designed an experience, not a channel — a coherent, risky bet.

Case & worked answers

Shell — A Company of Opportunity?

Shell's hub: 118 owners, 229 seekers, only 16 projects, none completed. Q5: two customer groups plus blocking line managers; demand-side failure; low similarity to current services/processes, unclear financial case; variability enters through unstructured briefs. Q6: SCAMPER redesign — mandatory brief template, link to performance reviews, short project packages, pre-approved time, no unsponsored postings, reverse to skill postings; measure projects per seeker, completion, time-to-match. Q7: durable capability lives in processes, incentives and culture, not one leader.

Practice questions

Try answering first. The answer key is below each question.

1. Starting from new technology then finding a market is…

A. Market-pull
B. Technology-push
C. Platform
D. Quick-build

Answer: B. Technology-push.

2. The tool translating weighted needs into technical targets is…

A. SCAMPER
B. House of Quality (QFD)
C. Value curve
D. Gantt

Answer: B. QFD.

3. Apple removing the floppy drive is SCAMPER's…

A. Substitute
B. Combine
C. Eliminate
D. Reverse

Answer: C. Eliminate.

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4. A VA/VE question is…

A. Can two parts be combined?
B. Share price?
C. Project lead?
D. Ad timing?

Answer: A. One of the four questions.

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5. Shell's hub's main problem was…

A. Too few seekers
B. Few projects and unstructured briefs
C. Cloud failed
D. Low NPS

Answer: B. 229 vs 16; NPS was 67.

Sources

- P3MD MK4 OSCM — Design of Products and Services (class slides, final); Design Thinking (STS)
- Jacobs & Chase — OSCM Ch. 3; HBS 9-609-066 (Apple); HBS 9-320-025 (Shell)